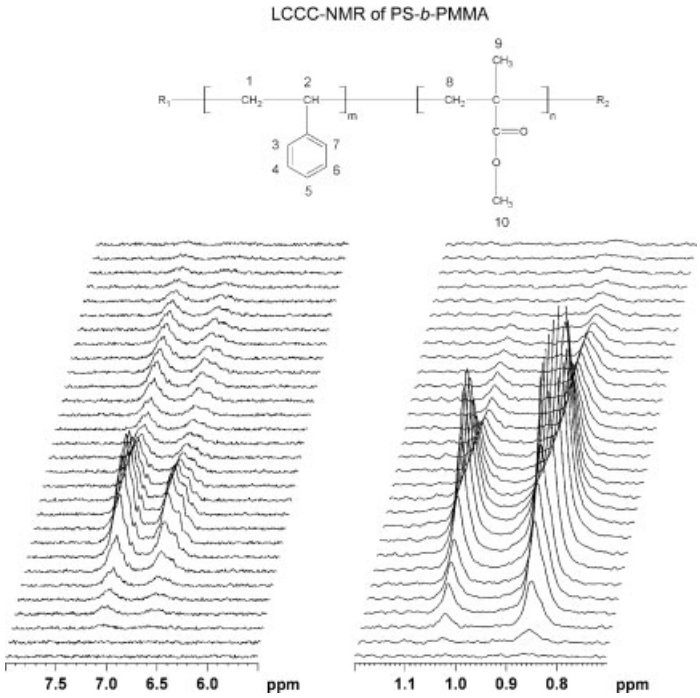
On-Line HPLC-NMR of PS-*b*-PMMA and Blends of PS and PMMA: LCCC-NMR at Critical Conditions of PS

Wolf Hiller,* Pritish Sinha, Harald Pasch

Blends and copolymers of PS and PMMA were analysed by LC coupled to ^{1}H NMR at the critical point of adsorption. The separation of the polymers was achieved at chromatographic conditions that correspond to the critical point of PS and the size-exclusion mode of PMMA. Copolymers and blends were analysed by on-line coupled ^{1}H NMR. For the homopolymer blends, separation into the components was achieved, while the copolymers were separated with regard to the block lengths of the PMMA blocks. The tacticity of the PMMA blocks could be determined as a function of molar mass by HPLC-NMR. This technique can deliver the true molar mass and the true chemical composition of the copolymers.



Introduction

The on-line coupling of HPLC and ^{1}H NMR is a powerful tool for the analysis of complex mixtures. However, so far this method was rarely used for the analysis of polymers. Most of the papers reported studies in a single mobile phase, e.g. deuterated chloroform or tetrahydrofuran (THF), where size exclusion chromatography (SEC) was used as the separation technique. Hatada et al. used SEC-NMR for the analysis of poly(methyl methacrylate) (PMMA).[1–4] He also analysed poly[(methyl methacrylate)-*co*-(butyl methacrylate)] copolymers by SEC-NMR with regard to the chemical composition.[5] Albert studied the chemical composition of poly[styrene-*co*-(butyl acrylate)] copolymers with SEC-NMR.[6] Up to now, there are only a few HPLC-NMR applications related to liquid adsorption chromatography (LAC) of polymers.[7–9] A mixture of technical poly(ethylene oxide)s was separated in acetonitrile (ACN)/D_2O with a reversed-phase (RP) stationary phase and the structures were assigned by LAC-NMR.[7] One of our own works was related to the separation of

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oligostyrenes by chain length and the subsequent analysis of the separated components by tacticity.[8] Krämer et al. reported on the separation and analysis of poly[styrene-*co*-(ethyl acrylate)] copolymers by gradient HPLC-NMR.[9] In this case, the chemical composition distribution (CCD) was determined in a chromatographic system comprising a mobile phase of protonated ACN/THF and a reversed stationary phase.

In general, the analysis of copolymers is rather complicated due to the simultaneous distributions in molar mass and chemical distribution. It has been shown that the concept of liquid chromatography at critical conditions (LCCC) can be applied for the separation and analysis of block copolymers.[10–14] While operating at the critical conditions of one block component, which is then considered to be 'chromatographically invisible', the other block component can be analysed, e.g. in the SEC mode. Further papers reported the analysis of block copolymers using LCCC. Pasch et al. analysed the blocks of poly(decyl methacrylate)-*block*-poly(methyl methacrylate) at critical conditions.[15] Falkenhagen et al. characterised both blocks of poly(methyl methacrylate)-*block*-poly(*tert*-butyl methacrylate) by operating at the critical conditions for the corresponding homopolymers.[16] Polystyrene-*block*-polyisoprene copolymers were analysed by Lee et al.[17] using temperature-gradient interaction chromatography (TGIC). In this case, the critical conditions of the blocks were adjusted by using only a single solvent and varying the column temperature.

Only few papers have been published on the coupling of LCCC and NMR. Kitayama et al. reported the analysis of poly(ethyl methacrylate)s regarding tacticity.[18] Pasch et al. studied the functionality-type distribution of poly(ethylene oxide)s by LCCC-NMR.[19–20]

However, up to now there are no applications of LCCC-NMR of block copolymers. It will be the main focus of this paper to analyse polystyrene-*block*-poly(methyl methacrylate) (PS-*b*-PMMA) copolymers and blends of PS and PMMA by means of LCCC-NMR at critical conditions of PS. The separation of the copolymers and blends is based on a chromatographic method that has been published earlier by Pasch et al.[21] LCCC-NMR will be used to correlate the chemical composition and the microstructure of the copolymers and the polymer blends to molar mass. In particular, quantitative information on the tacticity of the PMMA blocks will be given that is not accessible by other methods.

Experimental Part

Samples

PS and PMMA calibration standards as well as the PS-*b*-PMMA copolymers were produced by PSS GmbH (Mainz, Germany). The calibration standards had the following weight-average molar masses ($\overline{M}_w$): PS 4.05, 15, 35, 65, 145, 470 $kg \cdot mol^{-1}$; PMMA 1.9, 10.9, 29, 84.9, 253, 640 $kg \cdot mol^{-1}$. The molar masses of the copolymers and polymer blends are summarised in Table 1.

Table 1. Molar masses of the block copolymers and the blends of homopolymers as given by the supplier.

Sample	$\overline{M}_w$ PS-*b*-PMMA	Sample	$\overline{M}_w$ PS/$\overline{M}_w$ PMMA
	$kg \cdot mol^{-1}$		$kg \cdot mol^{-1}$
1	20.5	7	65/1.9
2	65	8	65/10.9
3	108	9	65/29
4	158	10	65/84.9
5	610	11	65/253
–	–	12	65/640

HPLC

An HPLC system Agilent 1100 equipped with a UV detector was used for the critical chromatography experiments. The chromatographic separations were conducted on Macherey-Nagel Nucleosil C_{18} 300-5 and 1000-7 with column sizes of 250×4 mm inner diameter. The concentration of the injected samples was 0.5 $mg \cdot mL^{-1}$ for determining the critical conditions and 10 μL were injected. For the HPLC-NMR measurements, the concentration was 20 $mg \cdot mL^{-1}$ for the copolymers and 6 $mg \cdot mL^{-1}$ for the blends; injection volume in this case was 50 μL. Normal HPLC grade ACN and THF were used as the mobile phase at a flow rate of 0.5 $mL \cdot min^{-1}$.

NMR

The NMR experiments were executed on a 400 MHz spectrometer AVANCE (Bruker BioSpin GmbH, Rheinstetten, Germany). The measurements were performed with a flow probe containing a 60 μL flow cell. The probe was a $^1H\{^{13}C\}$ inverse detection probe equipped with a shielded pulsed field-gradient coil. The gradient strength was 53 $G \cdot cm^{-1}$. The 90° ^1H pulse was 4.7 μs. WET solvent suppression[22] was applied to ACN and THF. Three frequencies were suppressed. On-flow experiments were performed for studying the critical conditions. Twelve scans per free induction decay (FID) were acquired with an acquisition time of 1.1 s (16 kb data points) and a relaxation delay of 0.1 s.

Results and Discussion

The goal of the present paper is the development of an analytical procedure for the analysis of PS-*b*-PMMA copolymers and the corresponding homopolymer blends with regard to CCD and microstructure as a function of molar mass distribution (MMD). On-line coupled

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HPLC-^{1}H NMR shall be used as the analytical technique. One of the major benefits of HPLC-NMR is that the chemical composition and the microstructure of the monomer units can be quantified in dependence of the retention time without any calibration.

As has been shown previously, LCCC is a suitable tool for separating block copolymers regarding the different block components. In general, when selecting appropriate eluents for LCCC, the main attention is being paid to evaluating the polarities of both the stationary and mobile phases in comparison to the polarities of the monomer units. However, if LCCC should also be coupled to NMR, then another problem has to be taken into account. The positions of the solvent signals relative to the signals of the analytes have to be considered. It is important that both monomer units of the block copolymers can be detected via proton signals that do not overlap with the mobile phase signals. Figure 1 shows the ^{1}H NMR spectra of a PS-*b*-PMMA copolymer dissolved in a mixture of ACN and THF without adding deuterated solvents [Figure 1(a)] and in deuterated dichloromethane [Figure 1(b)]. Solvent suppression is applied to the signals of both ACN and THF. It is evident from Figure 1 that the α-CH_3 group of the PMMA block and the aromatic protons of the PS block are unaffected by the solvent suppression. These signals can be used for the determination of the CCD. Figure 1 also indicates that the tacticity of the PMMA block can be calculated from the α-CH_3 group. The syndiotactic, heterotactic and isotactic triads are represented by rr, mr and mm, respectively.

For a first overview on the tacticity of the total samples, the PMMA calibration standards and the PS-*b*-PMMA block copolymers were measured and the tacticity compositions were determined. The corresponding results are summarised in Table 2.

The solvent combination of ACN/THF is well suited to be used as a mobile phase in HPLC-NMR experiments. It has already been shown in[21] that ACN and THF are suitable eluents for establishing critical conditions of PS. Using a reversed stationary phase, e.g. silica gel C_{18}, PMMA will

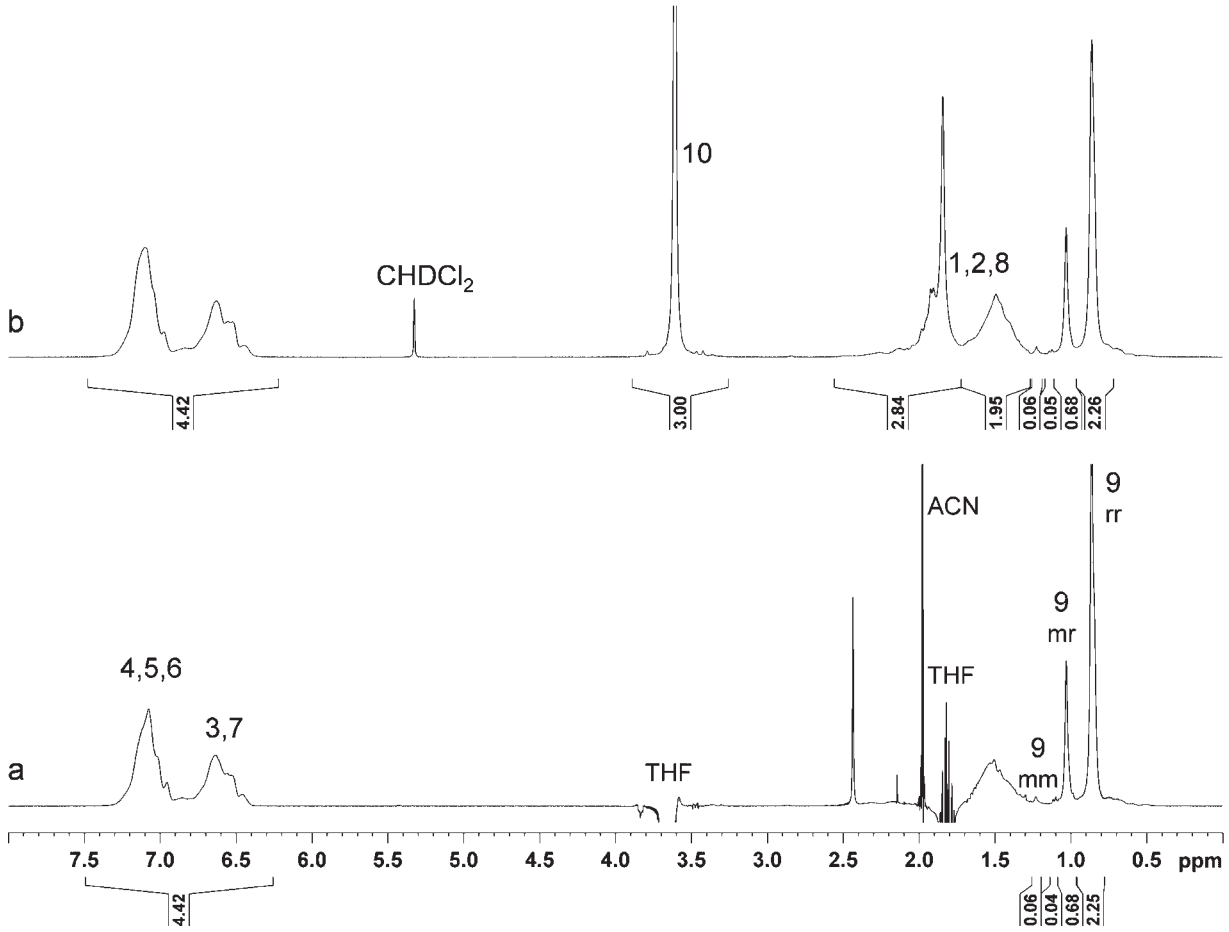

Figure 1. ^{1}H spectrum of a PS-*b*-PMMA copolymer (sample 1) in (a) nondeuterated ACN/THF with WET solvent suppression and (b) in CD_2Cl_2. The tacticity represented by the syndiotactic, heterotactic and isotactic triads of the α-CH_3 group is indicated by rr, mr and mm, respectively. The assignments are given according to Scheme 1.

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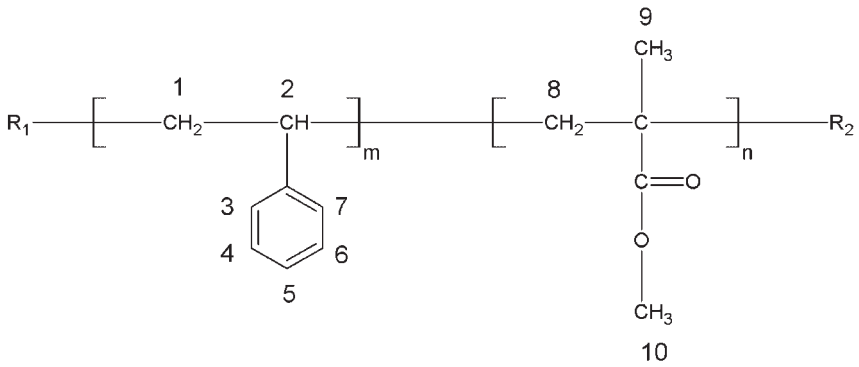

Scheme 1. Structure of PS-*b*-PMMA.

elute in the SEC mode, while PS is eluting at critical conditions. Figure 2 shows the critical diagram for using Nucleosil RP-18 columns at ambient temperature (22 °C). The critical conditions correspond to a mobile phase composition of THF–ACN of 48:52 v/v in agreement with ref.[21]

These critical conditions are used for the analysis of blends of PS and PMMA as well as for PS-*b*-PMMA copolymers. Figure 3 shows the HPLC-NMR on-flow run of a blend of PS, PMMA and the PS-*b*-PMMA block copolymer sample 5. The on-flow run is presented as a contour plot, which is processed as a two-dimensional NMR spectrum. Fourier transformation is performed on the series of FIDs and displayed as the contour plot of the retention time versus chemical shift. According to Figure 3, the block copolymer elutes first (retention time (RT) = 8.1 min) (a), followed by PMMA (RT = 9.9 min) (b) and PS (RT = 12.7 min) (c). Since critical conditions of PS are chosen, PS is eluting as the last component. The other components are eluting in the SEC mode. Therefore, the block copolymer with the larger PMMA block is eluting first followed by the PMMA homopolymer. Instead of presenting a contour plot, Figure 3 could also be shown as a stacked plot, where each spectrum is displayed separately with its individual signal intensities. These spectra would correspond to the traces of the contour diagram. The projections towards the elution time axis present the concentration profiles of the styrene (full line) and MMA units (dotted line).

Figure 4 presents the most intense traces of the separated components. Figure 4(a) shows the typical chemical shifts of the copolymer as indicated in Figure 1. However, the OCH_3 group of the PMMA block overlaps with THF and is suppressed. The second eluting peak can be assigned to PMMA ($\overline{M}_w = 29\,000$) [Figure 4(b)]. The homopolymer is mainly represented by the α-CH_3 groups at 0.8–1.1 ppm. This trace also contains some residual aromatic signals belonging to the copolymer. The OCH_3 group is again suppressed. The third trace belongs to PS ($\overline{M}_w = 65\,000$) and is represented by the aromatic signals at 6.4–7.3 ppm and the aliphatic region around 1.5 ppm.

Figure 5 shows the on-flow runs of all six blends of Table 1 at critical conditions. In this case, all spectral regions are displayed. The aromatic protons of PS and the α-CH_3 groups of PMMA can be observed. These on-flow runs can be used to produce NMR chromatograms by taking the vertical projections of both regions together. Figure 6 shows these projections. They verify the critical point of adsorption of PS at 12.7 min. As for PMMA, a SEC mode is indicated. The higher the molar mass of PMMA, the earlier is the elution of this homopolymer. Thus, it can

Table 2. Tacticity of the PMMA homopolymers and the PMMA blocks of the copolymers.

$\overline{M}_w$ PMMA	mm	mr	rr	Sample	mm	mr	rr
kg · mol^{-1}	%	%	%		%	%	%
1.9	11.3	39.5	49.2	1	2.4	23.0	74.6
10.9	4.7	36.8	58.5	2	1.7	21.1	77.2
29	4.8	36.6	58.6	3	3.0	26.1	70.9
84.9	4.0	48.0	48.0	4	1.4	20.7	77.9
253	1.0	19.7	79.3	5	2.3	24.3	73.4
640	4.6	49.7	45.7	–	–	–	–

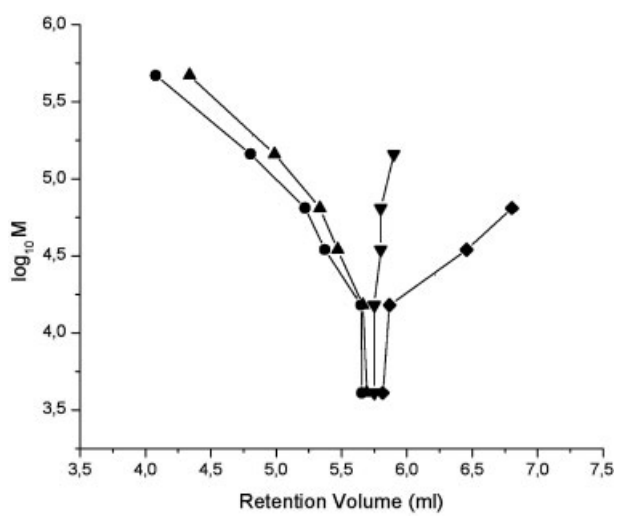

Figure 2. Critical diagram of molar mass *M* versus retention volume of PS collected with UV, stationary phase: RP-18 Nucleosil 300 + 1000; mobile phase THF/ACN • = 50:50, ▲ = 49:51, ▼ = 48:52, ◆ = 47:53 v/v.

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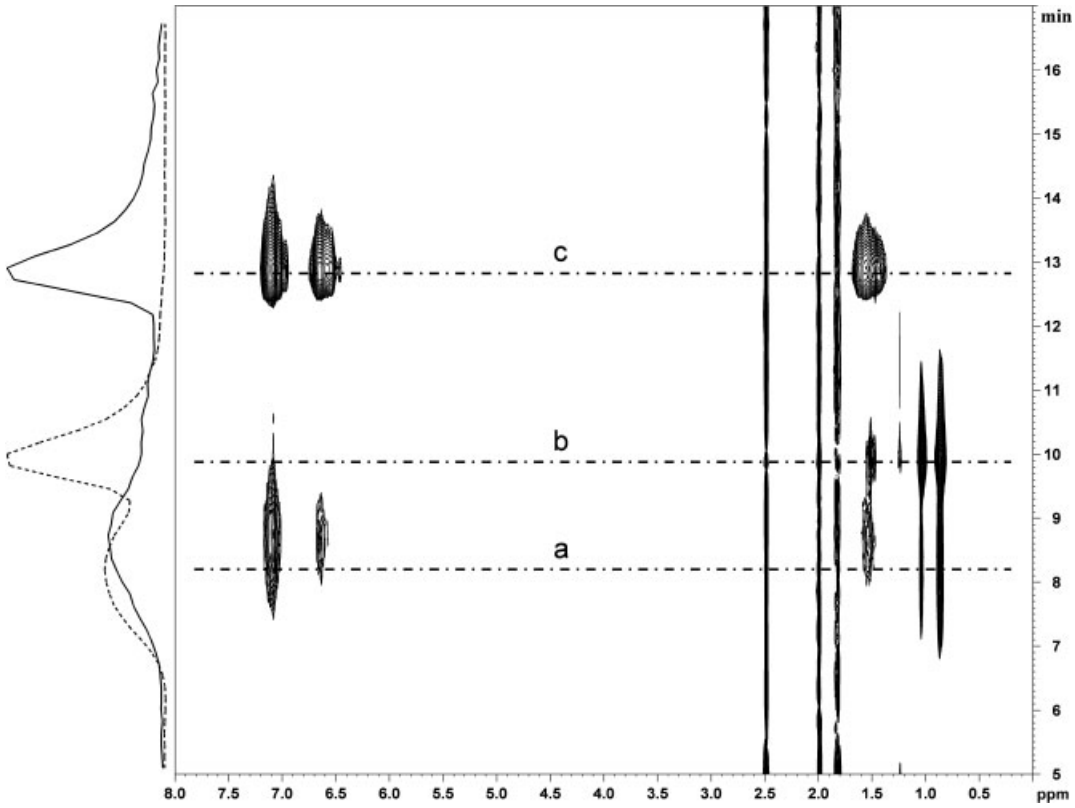

Figure 3. LCCC-NMR (400 MHz) on-flow run of the blend of PS-*b*-PMMA copolymer (sample 5) (a), PMMA ($\overline{M}_w = 29\,000$) (b) and PS ($\overline{M}_w = 65\,000$) (c) at the critical point of adsorption of PS; stationary phase: RP-18 Nucleosil 300 + 1000; mobile phase THF/ACN 48:52 v/v; flow rate 0.5 mL · min^{-1}; sample concentration 5 mg · mL^{-1} for each polymer; injection volume 50 μL. Projections taken from the vertical traces (solid line = aromatic region, dashed line = α-CH_3 group).

be concluded that these conditions are very useful for the separation of blends of PMMA and PS. Macko et al. have shown another method of LCCC for the separation of these blends at critical conditions of PS by using only a single solvent.[23] This method would probably also be useful for NMR.

Figure 7(a–e) shows the HPLC-NMR on-flow runs of all block copolymers presented in Table 1. It is evident from most of these figures that a separation into two different species takes place. The first component elutes at different retention times while the second component elutes near the critical retention time of PS. The identification of the two components can be done by assigning the observed chemical shifts. Since the first eluting peak shows coexisting signals at the aromatic and aliphatic regions, this peak can be assigned to the copolymer. The later eluting peak can be assigned to PS homopolymer based on the typical NMR chemical shifts of PS and the critical elution time of PS. This assignment is also in agreement with the vertical projections of the on-flow runs. While the solid line represents the distribution of the aromatic protons that are present both in the copolymer and the PS homopolymer, the dashed line represents the distribution of the α-CH_3 groups only due to the copolymer. Furthermore, it can be seen that the elution time of the copolymer peak decreases with increasing molar mass, which is in agreement with the SEC mode for the PMMA block. Consequently, the molar mass of the PMMA blocks in the copolymers can be determined by using the HPLC-NMR on-flow data of PMMA calibration standards. Figure 8 shows the vertical projections of all on-flow runs of the PMMA standards at critical conditions of PS. The vertical projections are taken from the proton intensity distributions of the α-CH_3 groups. Similar to standard chromatographic procedures, the determination of the molar mass versus elution volume function can be derived. Using the calibration curve given in Figure 9, the average molar masses of the PMMA blocks in the block copolymers were determined.

For the evaluation of the LCCC-NMR data, the block copolymer samples are also analysed by conventional SEC and by off-line ^{1}H NMR. The conventional SEC data are obtained in THF using cross-linked polystyrene (PS) as the stationary phase. The calibration is based on PS. As can be

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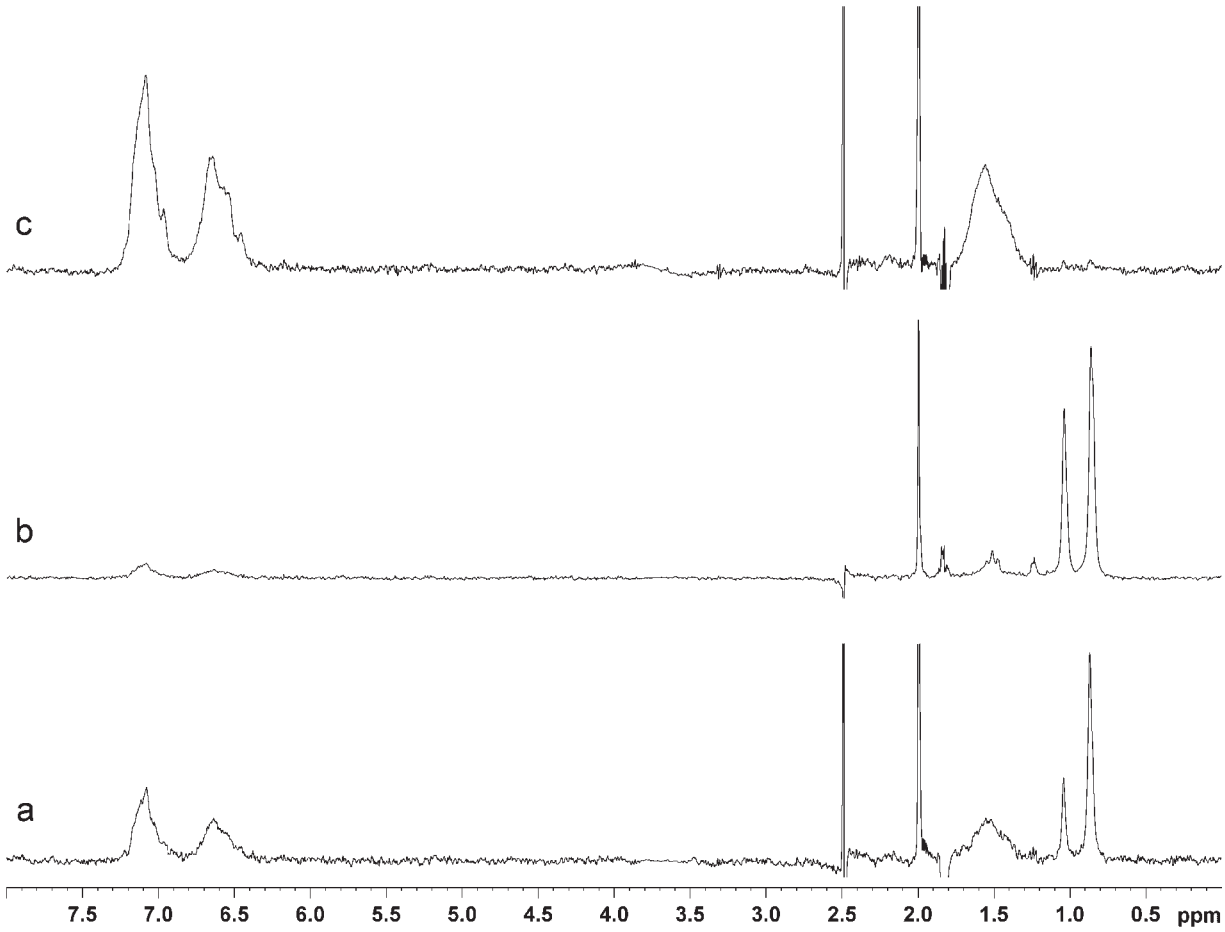

Figure 4. ^{1}H traces of the on-flow run of Figure 3: (a) PS-*b*-PMMA (sample 5, RT = 8.1 min); (b) PMMA ($\overline{M}_w$ = 29 000, RT = 9.9 min); (c) PS ($\overline{M}_w$ = 65 000, RT = 12.7 min).

seen from Table 3, the SEC molar mass data are in good agreement with the nominal molar masses given by the producer. Off-line ^{1}H NMR is used to determine the chemical composition of the samples. From the signal intensities of the aromatic protons (styrene) and the α-CH_3 protons (methyl methacrylate) the average composition of the block copolymers can be calculated, see Table 3. One has to keep in mind that the numbers given here are related to the total samples. These contain, in addition to the true block copolymer, certain amounts of PS homopolymer.

In the next step the molar masses of the PMMA blocks are calculated from SEC (M_p values at peak maximum) and off-line NMR data. The calculation was done using the following equation

$$M_{copol} = N_{total}[x_{mol}M_S + (1 - x_{mol})M_{MMA}] \quad (1)$$

where M_{copol} is the molar mass of the copolymer, N_{total} the number of all monomer units of the copolymer chain, x_{mol} the molar content of styrene units, M_S the molar mass of one styrene unit (104 g · mol^{-1}) and M_{MMA} the molar mass of one methyl methacrylate unit (100 g · mol^{-1}). Accordingly, N_{total} can be calculated using the copolymer molar mass and the number of monomer units

$$N_{total} = \{1/[x_{mol} + (1 - x_{mol})/1.04]\}M_{copol}/M_S \quad (2)$$

$$N_{total} = N_S + N_{MMA} \quad (3)$$

where N_S and N_{MMA} are the total numbers of styrene and MMA units in the copolymer chain, respectively. The molar mass of the PMMA block is then obtained as follows:

$$M_{PMMA} = N_{MMA}M_{MMA} \quad (4)$$

The calculated molar masses of the PMMA blocks are summarised in Table 4 and compared to the molar masses that were obtained by LCCC-NMR. In the case of LCCC-NMR,

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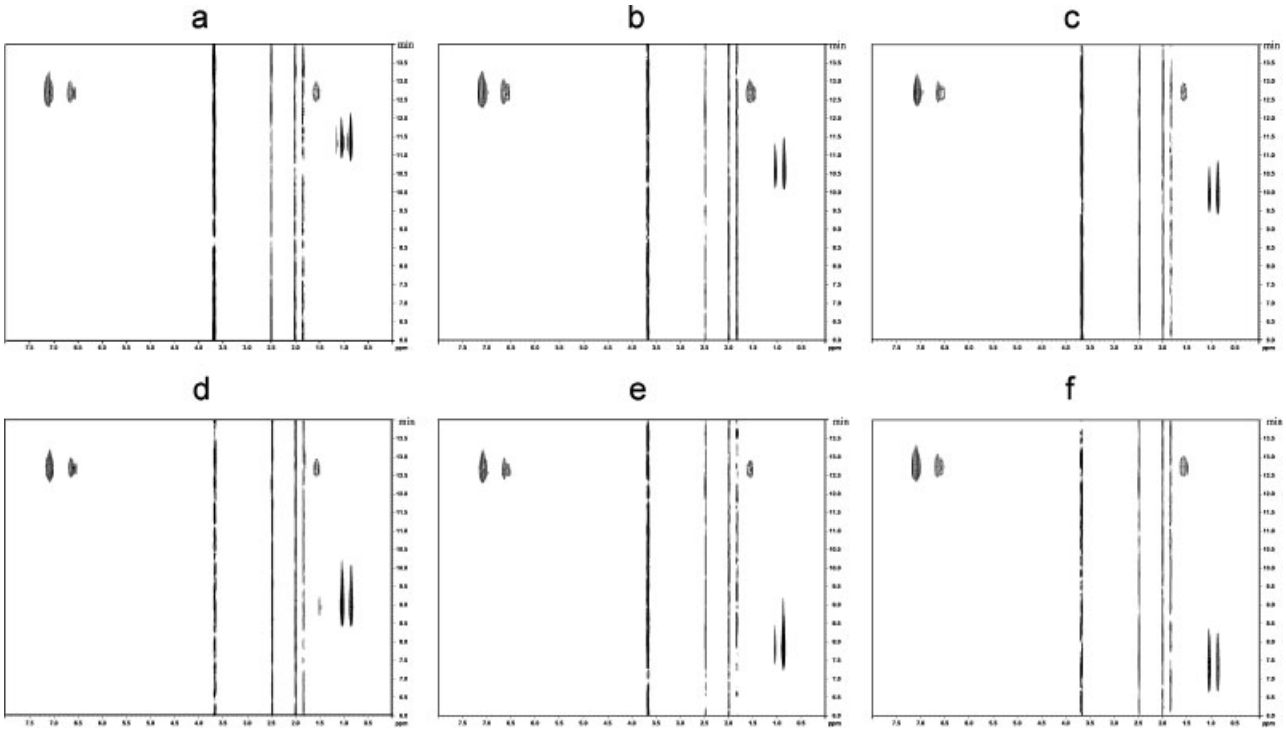

Figure 5. LCCC-NMR (400 MHz) on-flow runs of six blends of PS and PMMA of different molar masses; experimental conditions see Figure 3; samples 6 (a), 7 (b), 8 (c), 9 (d), 10 (e), 11 (f). Concentration was 3 mg · mL^{-1} for each polymer; injection volume was 50 μL.

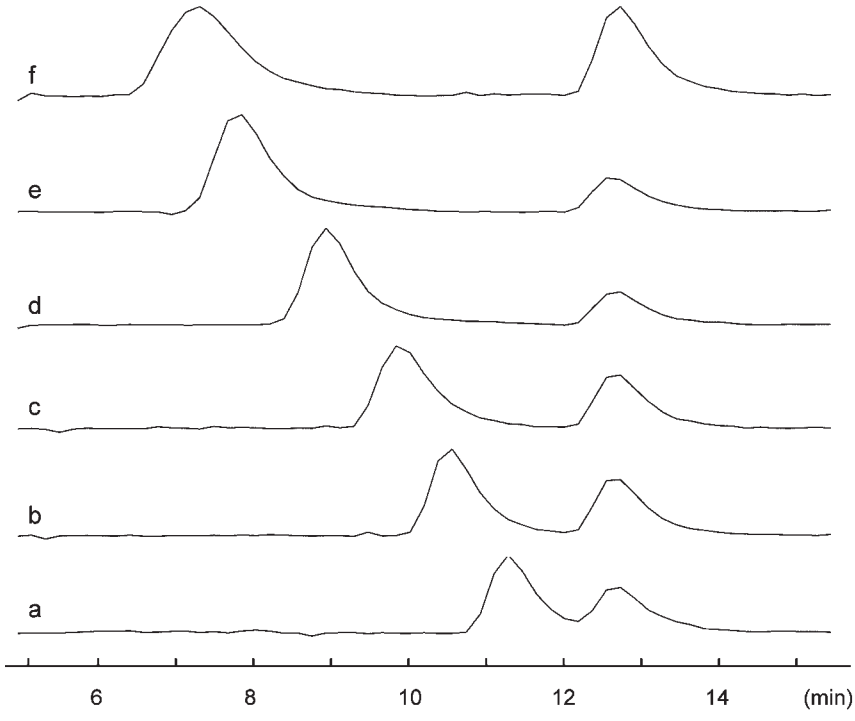

Figure 6. Vertical projections of the on-flow runs of Figure 5 (labels as in Figure 5).

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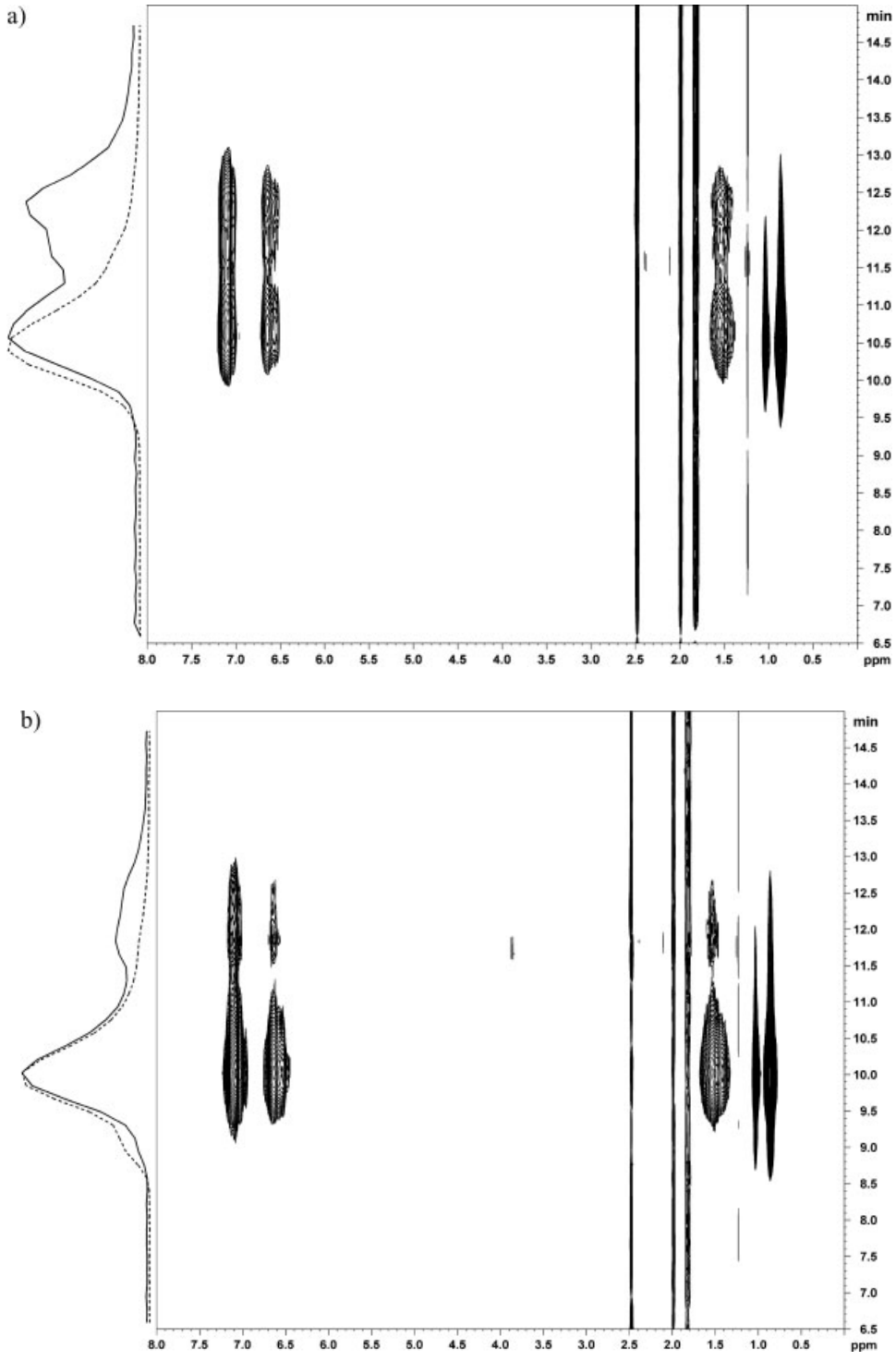

Figure 7. LCCC-NMR (400 MHz) on-flow runs of the PS-*b*-PMMA copolymers; experimental conditions see Figure 3; vertical projections are taken from the aromatic (solid line) and α-CH_3 proton signals (dashed line); samples 1 (a), 2 (b), 3 (c), 4 (d), 5 (e). Concentration was 20 mg · mL^{-1} for each copolymer; injection volume was 50 μL.

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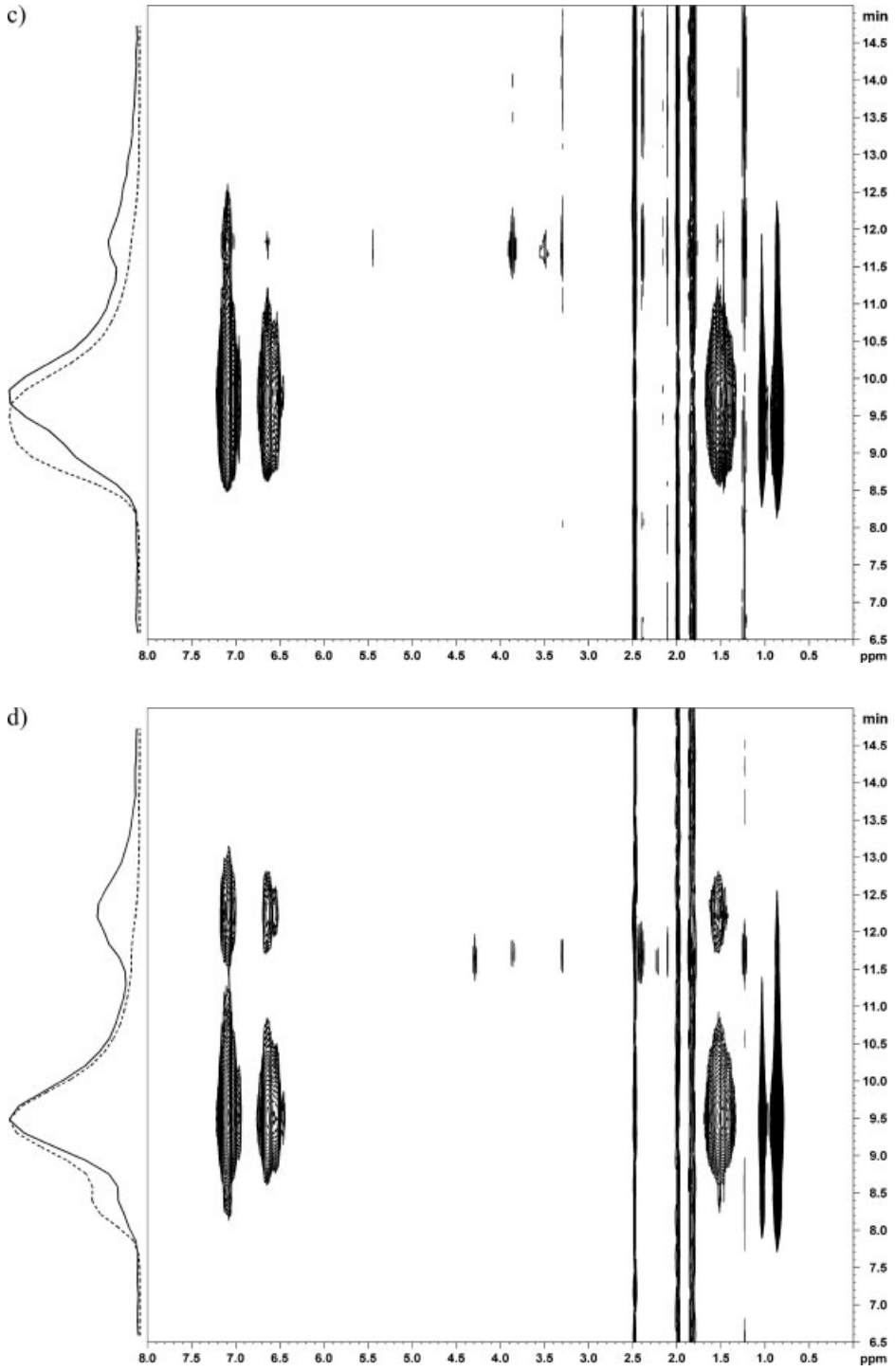

Figure 7. (continued)

the maximum intensity of the α-CH_3 signals was used for determining the retention time. The calculation of the molar mass is based on the fitting of Figure 9.

Considering the complexity of the experiments and the size of the NMR flow cell, the PMMA molar masses obtained by both approaches agree quite well. There are different sources of error that can affect the results. First of all, the accuracy of establishing the critical conditions will influence the results. A second source of error is the accuracy of the molar mass analysis by off-line SEC. A PS

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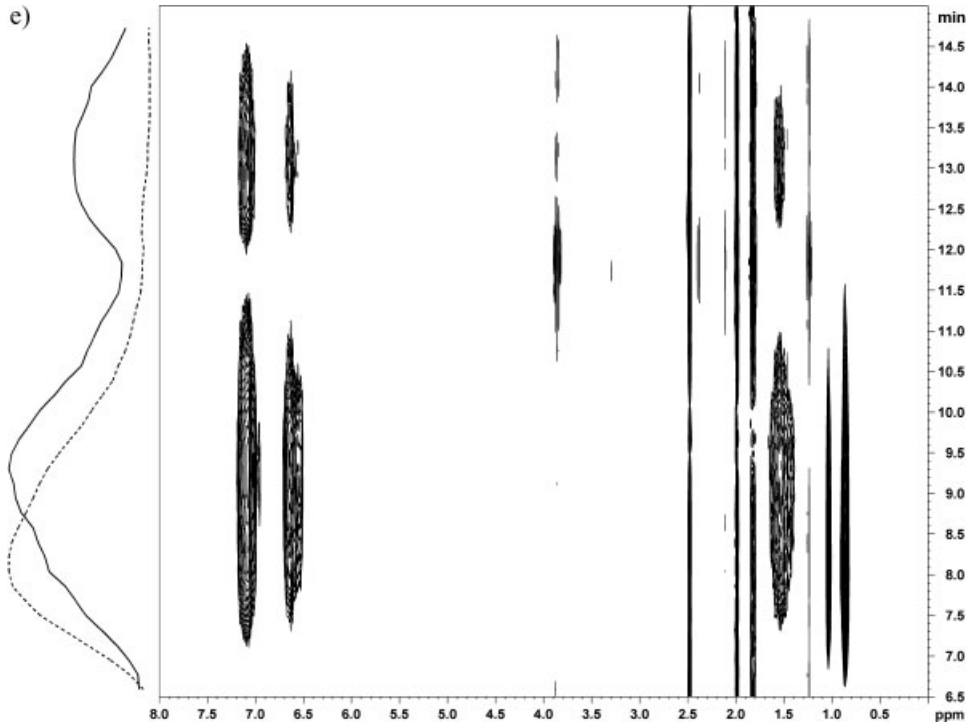

Figure 7. (continued)

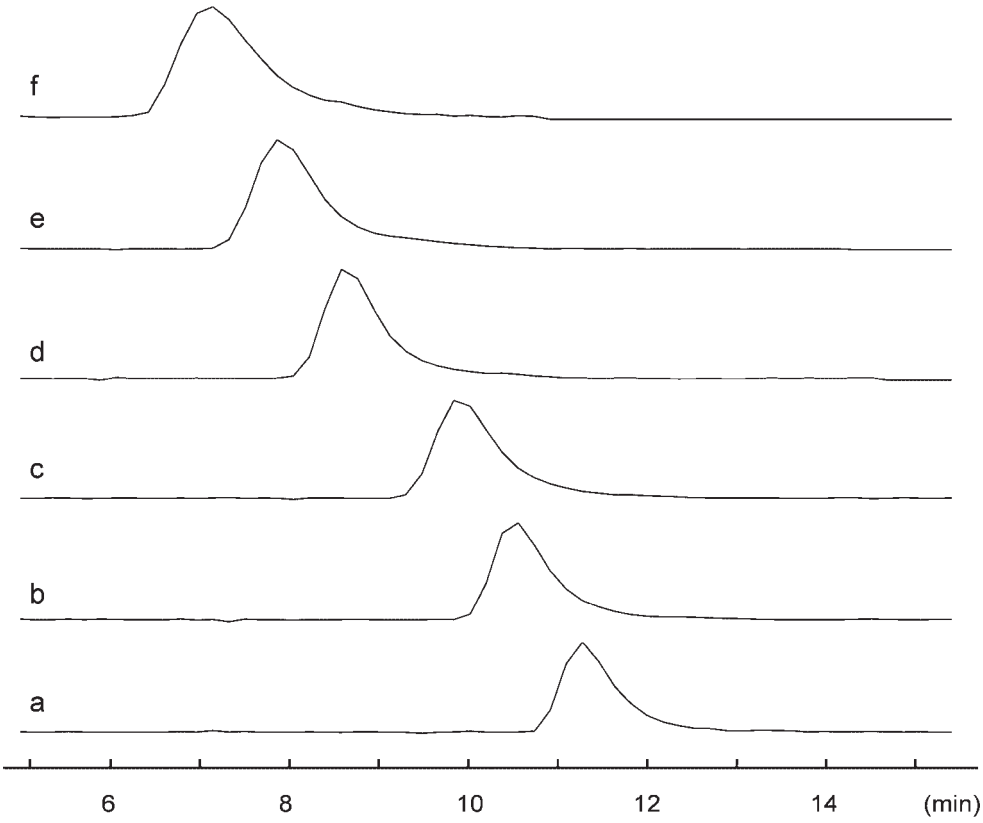

Figure 8. Vertical projections of the LCCC-NMR (400 MHz) on-flow runs of six PMMA calibration standards; experimental conditions see Figure 3; samples (a) $\overline{M}_w = 1\,900$; (b) $\overline{M}_w = 10\,900$; (c) $\overline{M}_w = 29\,000$; (d) $\overline{M}_w = 84\,900$; (e) $\overline{M}_w = 253\,000$; (f) $\overline{M}_w = 640\,000$.

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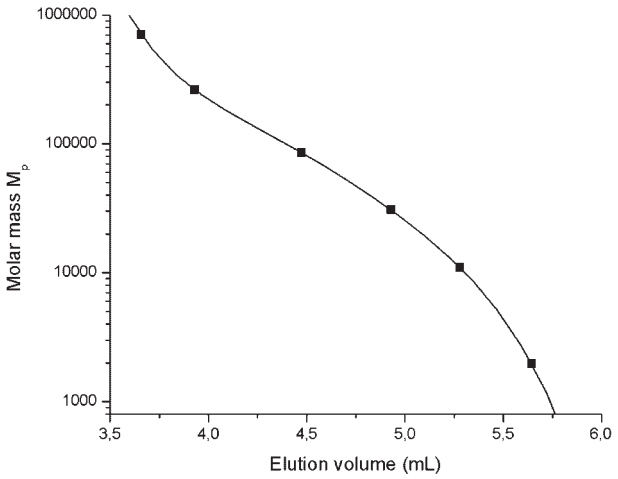

Figure 9. Calibration curve of molar mass versus retention volume for PMMA, ■ = data point based on the vertical projections from Figure 8, solid line = fitted curve with fifth polynomial.

calibration was used to determine the molar masses of the samples. In addition, SEC does not separate the block copolymers from the detected PS homopolymers and, accordingly, the presence of these homopolymer fractions will affect the molar mass analysis. To the same extent this is true for the chemical composition that was analysed by off-line ^{1}H NMR. These numbers were obtained for the total samples and may deviate from the chemical composition of the true block copolymers.

The total volume of the NMR flow cell, which is about twice the active volume, can cause tailing during the elution and can finally result in spreading out the eluted peak. Nevertheless, the 60 μL flow cell is already a very good compromise between LC resolution and NMR sensitivity.

One of the significant benefits of HPLC-NMR is the ability to also determine the CCD of the copolymers without any standards. Using the on-flow data presented in Figure 7(a–e), it is possible to calculate the chemical composition

Table 4. Molar masses of the PMMA block of the copolymers determined by SEC and off-line NMR with Equation 4 or LCCC-NMR with the peak maximum of the α-CH_3 group.

Sample	$\overline{M}$ (PMMA block) by SEC and NMR	$\overline{M}_p$ (PMMA block) by LCCC-NMR
	kg · mol^{-1}	kg · mol^{-1}
1	9.1	14.2
2	30.4	27.1
3	61.9	48.1
4	51.8	48.1
5	196	196

at different elution volumes. Based on the calibration curve given in Figure 9, the CCD can be determined as a function of molar mass at critical conditions of PS. Figure 10 shows the plots of the CCD versus retention time for the different block copolymers. The figure also shows the NMR projections of the aromatic region (solid line, normalized to one proton) and the α-CH_3 group (dashed line, normalized to one proton). The following general conclusions can be derived from Figure 10(a–e):

(i) All samples show a bimodal MMD (indicated by the solid line). The distribution at lower retention times represents the copolymer and the higher retention time distribution belongs to PS homopolymer.

(ii) It was found that all block copolymers show a significant chemical heterogeneity. Samples 1, 3 and 5 [Figure 10(a, c, e)] are very broadly distributed with regard to CCD while samples 2 and 4 [Figure 10(b, d)] show a more moderate dependence of the CCD versus retention time in the region of the copolymer. The more pronounced chemical heterogeneity of samples 1, 3 and 5 may have two reasons: (A) the PMMA block in the block copolymers may have a higher polydispersity or (B) the chemical composition of the block copolymer may exhibit a larger drift.

Table 3. SEC and ^{1}H NMR analyses of the block copolymers.

Sample	Nominal $\overline{M}_w$	$\overline{M}_w$ by SEC	$\overline{M}_n$ by SEC	Styrene/MMA by NMR
	kg · mol^{-1}	kg · mol^{-1}	kg · mol^{-1}	mol-%
1	20.5	20.5	17.6	47.0/53.0
2	65	75	69	45.1/54.9
3	108	132	121.5	48.1/51.9
4	158	164	100	47.4/52.6
5	610	668	571	64.8/35.2

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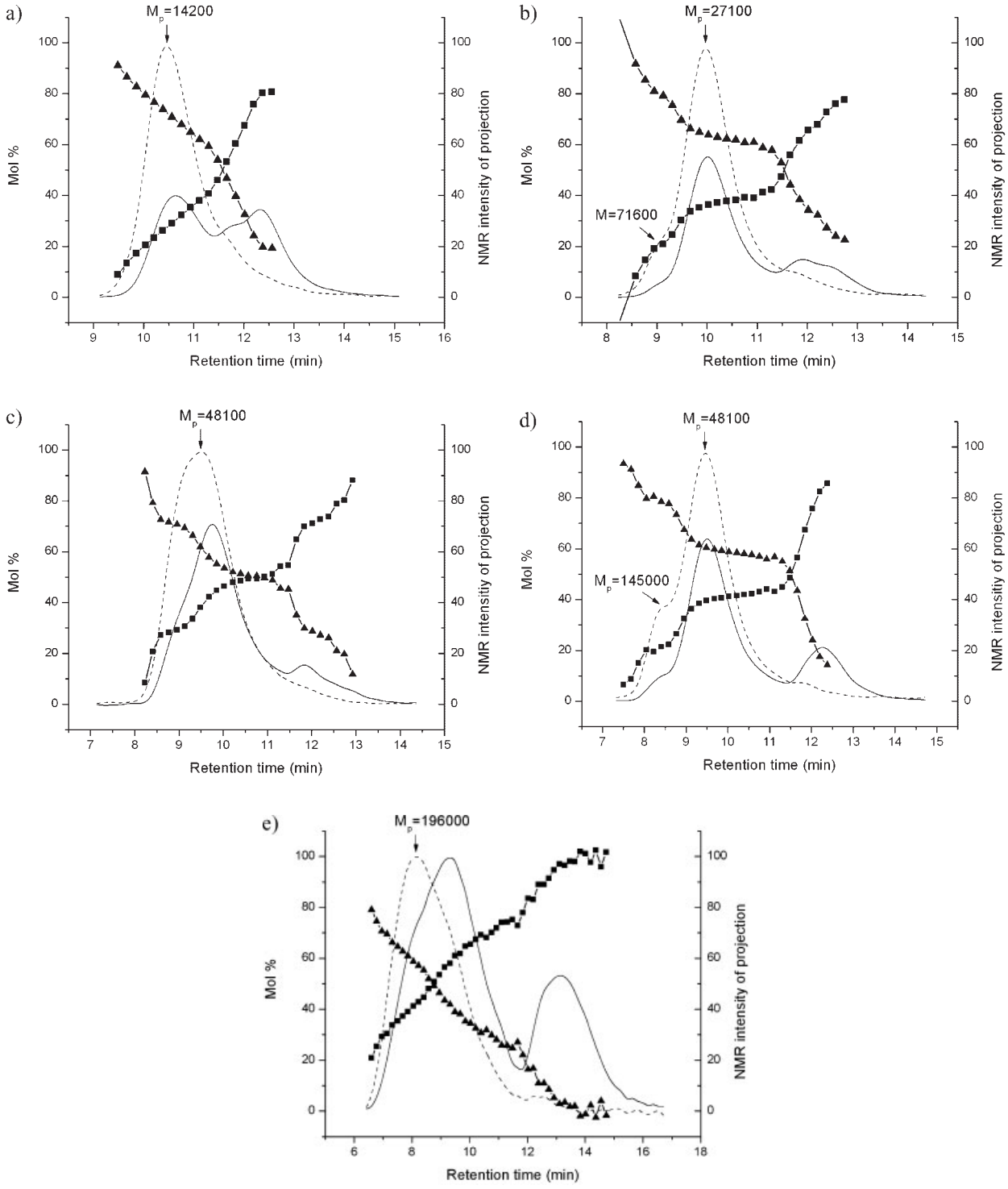

Figure 10. Chemical composition of PS-*b*-PMMA copolymers versus molar mass calculated from Figure 7 and 8 (■ = mol% PS, ▲ = mol% MMA), dashed line = NMR projection of the α-CH_3 group, solid line = NMR projection of the aromatic region (both taken from Figure 7), samples 1 (a), 2 (b), 3 (c), 4 (d), 5 (e)

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Table 5. True molar masses of the PMMA block of the copolymers, true copolymer chemical compositions and calculated total molar masses of the block copolymers.

Sample	$\overline{M}_p$ (PMMA block) by LCCC-NMR	Styrene/MMA of copolymer by LCCC-NMR	$\overline{M}_p$ of block copolymer
	$kg \cdot mol^{-1}$	mol-%	$kg \cdot mol^{-1}$
1	14.2	30.0/70.0	20.5
2	27.1	34.8/65.2	42.1
3	48.1	40.3/59.7	81.9
4	48.1	38.4/61.6	79.3
5	196	52.2/47.8	418.6

(iii) In general, it was found that the high molar mass region (low retention times) is rich in PMMA, the lower molar masses (high retention times) are dominated by PS.

(iv) The retention time of the PMMA projection profiles (α-CH_3) can be used for calculating M_p of the PMMA block.

In addition to these general statements, further structural peculiarities can be obtained from the CCD plots. First of all, the plots indicate that all samples are mixtures of the block copolymer and PS homopolymer. Accordingly, the average chemical composition given in Table 3 does not correspond to the true chemical composition of the block copolymer fraction. As can be seen in Figure 10(a), the true block copolymer has a composition of roughly 70 mol-% MMA and 30 mol-% styrene. This is quite different from the average sample composition of 53:47 mol-% MMA/styrene.

The samples 2 and 4 show shoulders towards very low retention times that have a chemical composition different from the composition of the main copolymer fraction. In both cases, these shoulders indicate fractions that have a higher PMMA content. Assumingly, these fractions are the result of coupling reactions that are taking place during the living polymerisation. Coupling can take place between two living chain ends when a suitable coupling component is present in the reaction mixture as an impurity. The formation of coupling products is well documented for anionic polymerisation. The molar masses of the PMMA blocks of these coupling products are roughly triple the molar masses of the PMMA blocks in the diblock copolymers. This could indicate the formation of multiblock fractions.

Based on the data that are obtained by LCCC-NMR, the chemical structure and molar mass of the true block copolymer fractions of samples 1–5 can be determined. As has been shown from LCCC with PMMA calibration, the molar masses of the PMMA blocks of the block copolymers are obtained. It should be noted that for this analysis NMR

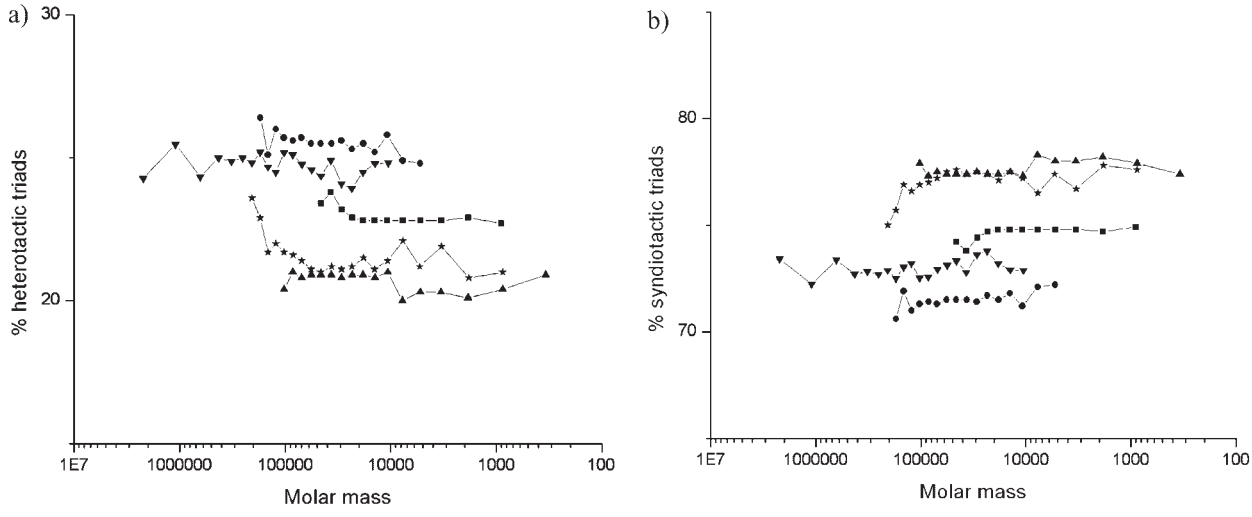

Figure 11. Tacticity of the PMMA block of PS-*b*-PMMA copolymers versus molar mass calculated from Figure 7 and 8, (a) heterotactic and (b) syndiotactic triads (■ = sample 1, ▲ = sample 2, ● = sample 3, * = sample 4, ▼ = sample 5).

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is a much more suitable detector than any concentration detector because it selectively detects the concentration profile of the MMA units instead of the total concentration profile. The true chemical compositions of the block copolymers are determined from the MMA and styrene concentration traces given in Figure 10. As can be seen in Table 5, the chemical compositions of the block copolymer fractions are significantly different from the data for the total samples.

Taking now the true PMMA block molar masses and the true copolymer compositions, the total molar masses of the block copolymers can be calculated, see Table 5. A comparison with the average molar masses of the total samples shows significant differences, compare Table 3 and 5.

The study of the block copolymers can also be done by using dual detection, e.g. ultraviolet (UV) and refractive index (RI). However, the advantage of the NMR is that it can be used as a concentration detector for both monomer units and the analysis of the microstructure.

Finally, the tacticity of the PMMA blocks shall be investigated. Due to the fact that the α-CH_3 groups in the ^{1}H NMR spectra reveal also the syndiotactic, heterotactic and isotactic triads (see Figure 1), the tacticity is determined from the on-flow data of Figure 7(a–e). The on-flow data, however, do not show the isotactic signals because of overlapping impurities. Therefore, the isotactic part was estimated by average data taken from Table 2 and supposed to be constant during the on-flow run. Figure 11 presents the calculated tacticities of the PMMA blocks as a function of the molar mass for the critical conditions of PS. It shows that the tacticity of the PMMA blocks is constant and does not depend on molar mass. The blocks are predominantly syndiotactic. This behaviour was found for all copolymers.

Conclusion

Finally, it can be concluded that HPLC-NMR is a versatile tool for the analysis of block copolymers. In particular, it is possible to determine the CCDs of PS-*b*-PMMA copolymers at chromatographic conditions that correspond to the critical point of the PS block. It was found that the PS-*b*-PMMA copolymers are chemically heterogeneous and contain fractions of PS homopolymer as well as copolymer fractions of different compositions. The molar masses of the PMMA blocks were determined by using a calibration procedure that is based on on-flow experiments. Moreover, it was found that the true chemical composition of the block copolymers can be determined by LCCC-NMR and can significantly differ from the average composition of the samples. Furthermore, the tacticity of the PMMA blocks was determined. Again, the on-flow experiments were used for calculating the syndio- and heterotactic triads as a function of the molar mass.

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[1] K. Hatada, K. Ute, Y. Okamoto, M. Imanari, N. Fujii, *Polym. Bull.* **1988**, *20*, 317.
[2] K. Hatada, K. Ute, M. Kashiyama, M. Imanari, *Polym. J.* **1990**, *22*, 218.
[3] K. Hatada, K. Ute, T. Kitayama, T. Nishimura, M. Kashiyama, N. Fujimoto, *Polym. Bull.* **1990**, *23*, 549.
[4] K. Ute, R. Niimi, M. Matsunaga, K. Hatada, T. Kitayama, *Macromol. Chem. Phys.* **2001**, *202*, 3081.
[5] K. Hatada, K. Ute, T. Kitayama, M. Yamamoto, T. Nishimura, M. Kashiyama, *Polym. Bull.* **1989**, *21*, 489.
[6] K. Albert, "*On-line LC-NMR and Related Techniques*", John Wiley & Sons Ltd., Chichester 2002.
[7] H. Pasch, W. Hiller, *Macromolecules* **1996**, *29*, 6556.
[8] H. Pasch, W. Hiller, R. Haner, *Polymer* **1998**, *39*, 1515.
[9] I. Krämer, W. Hiller, H. Pasch, *Macromol. Chem. Phys.* **2000**, *201*, 1662.
[10] H. Pasch, H. Much, G. Schulz, A. V. Gorshkov, *LC-GC Int.* **1992**, *5*, 38.
[11] H. Pasch, C. Brinkmann, H. Much, U. Just, *J. Chromatogr.* **1992**, *623*, 315.
[12] E. Gankina, B. Belenkii, I. Malakhova, *J. Planar Chromatogr.* **1991**, *4*, 199.
[13] T. M. Zimina, J. J. Kever, E. Y. Melenevskaya, A. F. Fell, *J. Chromatogr.* **1992**, *593*, 233.
[14] H. Pasch, C. Brinkmann, Y. Gallot, *Polymer* **1993**, *34*, 4100.
[15] H. Pasch, M. Augenstein, B. Trathnigg, *Macromol. Chem. Phys.* **1994**, *195*, 743.
[16] J. Falkenhagen, H. Much, W. Stauf, A. H. E. Müller, *Macromolecules* **2000**, *33*, 3687.
[17] W. Lee, D. Cho, T. Chang, K. J. Hanley, T. P. Lodge, *Macromolecules* **2001**, *34*, 2353.
[18] T. Kitayama, M. Janco, K. Ute, R. Niimi, K. Hatada, D. Berek, *Anal. Chem.* **2000**, *72*, 1518.
[19] G. Schlotterbeck, H. Pasch, K. Albert, *Polym. Bull.* **1997**, *38*, 673.
[20] W. Hiller, A. Brüll, D. Argyropoulos, E. Hoffmann, H. Pasch, *Magn. Res. Chem.* **2005**, *43*, 729.
[21] H. Pasch, Y. Gallot, B. Trathnigg, *Polymer* **1993**, *34*, 4986.
[22] S. H. Smallcombe, S. L. Patt, P. A. Keifer, *J. Magn. Res. A* **1995**, *117*, 295.
[23] T. Macko, D. Hunkeler, D. Berek, *Macromolecules* **2002**, *35*, 1797.

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